CS6230 — CAD for VLSI

QR Code Decoder Project

Physical Design and PPA Report

Post-Synthesis Design-Space Exploration (Table 5) and Place & Route of the Best Design (Table 6)

Student name	Kishore (Kishore Kotapati)
Roll number	EE26E016
Team partner	Rishi (EE25S082)
Department / programme	Electrical and Electronics Engineering, B.Tech
Institution	National Institute of Technology Manipur (B.Tech EEE)
Course	CS6230 — CAD for VLSI (CS6230)
Faculty / supervisor	[Faculty / course instructor — to be filled in]
Design	mkQRDecoder (Bluespec SystemVerilog), SAED 14nm RVT
Submission date	9 November 2026
Version	Final submission v1.0

Companion to **report.pdf** (RTL microarchitecture and verification).

Abstract

This report documents the physical-design (PD) work for the CS6230 QR-code decoder **mkQRDecoder**: five post-synthesis design-space-exploration (DSE) iterations with Synopsys Design Compiler (normal, non-topographical synthesis, SAED 14nm RVT, 40% external delay), and place-and-route of the best design with Synopsys IC Compiler (not ICC2), evaluated after PlaceOpt as announced by the course staff. All numbers are real results read from the tool reports; none are estimated.

Headline results. Table 5: the design was taken from 5.00 ns (200 MHz) to **1.20 ns (833.3 MHz)**, a **4.17×** faster clock, with zero violating paths in every run; the flop-to-flop critical path fell from 4.98 ns to 1.19 ns. Standard-cell area went from 60,138.2 to 55,777.5 µm² (-7.3%). Table 6: the best design (netlist *r4h12*, 1.20 ns) places with **0 violating setup and 0 violating hold paths**, **92.29% utilisation** (requirement ≥ 70%), a **243.98 × 243.45 µm** block, **52,285.97 µm²** placed cell area and **7.3985 mW** total power.

Keywords: QR decoder, Bluespec, Reed–Solomon, Design Compiler, IC Compiler, SAED14nm, PPA, design-space exploration, PlaceOpt.

Table of Contents

Chapter 1 Introduction . . . 6

1.1 Background and problem statement . . . 6

1.2 Objectives of this report . . . 6

1.3 Specification requirements that this report answers . . . 6

1.4 Division of work . . . 6

1.5 Organisation of the report . . . 6

Chapter 2 Tools, Libraries and Environment . . . 7

2.1 Software and technology . . . 7

2.2 Constraints (SDC) . . . 7

Chapter 3 Synthesis Flow with Design Compiler . . . 8

3.1 Flow overview . . . 8

3.2 Commands used for the DC runs . . . 8

3.3 Reports produced per run . . . 8

Chapter 4 Post-Synthesis Results: Design-Space Exploration (Table 5) . . . 11

4.1 Table 5 — Post-synthesis QoR data (official) . . . 11

4.2 Complete timing table — every clock period run (ns) . . . 11

4.3 What changed in each iteration (Design → Synthesis → Analyse → Re-design) . . . 13

4.4 Derived metrics: formulas and values . . . 14

Chapter 5 Place and Route of the Best Design (Table 6) . . . 14

5.1 Choice of the best design . . . 14

5.2 Every candidate period down to 0.55 ns and why it was or was not chosen . . . 14

5.3 Floorplan and flow settings . . . 15

5.4 Commands used for place and route . . . 15

5.5 Table 6 — Post-place QoR data (official) . . . 15

5.6 Best-design data sheet (all details) . . . 16

5.7 Core-size robustness of the chosen design . . . 18

Chapter 6 Auxiliary Experiments . . . 18

6.1 Frequency sweeps of earlier designs . . . 18

6.2 R5: a 4-bank store (experimental) . . . 18

Chapter 7 Analysis and Discussion . . . 19

7.1 DC versus ICC for the same netlist . . . 19

7.2 Findings of the design-space exploration . . . 20

7.3 Checks performed on the physical-design results . . . 20

7.4 Compliance with the specification . . . 20

Chapter 8 Problems Met and How They Were Resolved . . . 21

Chapter 9 Conclusion and Future Work . . . 21

Appendix A Command Reference . . . 21

Appendix B Screenshot Evidence Index . . . 22

Appendix C Submission Folder Inventory (pd/) . . . 23

Appendix D Requirement Traceability Matrix (physical design) . . . 23

Appendix E Glossary and Abbreviations . . . 24

Appendix F References . . . 24

List of Figures

Figure 2.1 — The SDC file and the library set-up of the DC script [SS-P01] . . . 7

Figure 3.1 — Terminal: the DC run of the best design (r4h12, 1.2 ns) [SS-P02] . . . 8

Figure 3.2 — Design 1: QoR report [SS-P03] . . . 9

Figure 3.3 — Design 2: QoR report [SS-P04] . . . 9

Figure 3.4 — Design 3: QoR report [SS-P05] . . . 9

Figure 3.5 — Design 4: QoR report [SS-P06] . . . 9

Figure 3.6 — Design 5 (best): QoR report [SS-P07] . . . 9

Figure 3.7 — Design 5: report_power [SS-P08] . . . 10

Figure 3.8 — Design Vision: logical hierarchy of mkQRDecoder (r4h12) [SS-P16] . . . 10

Figure 3.9 — Design Vision: top-level schematic symbol of mkQRDecoder [SS-P17] . . . 10

Figure 3.10 — Design Vision: schematic with all bus bits expanded [SS-P17b] . . . 10

Figure 3.11 — Design Vision: report_timing on the saved DC result [SS-P18] . . . 11

Figure 4.1 — Table 5 results: frequency, DC power and DC design area for the five iterations . . . 12

Figure 4.2 — DC total power against clock period for every passing run (circles: full decoder variants; squares: experimental R5) . . . 13

Figure 5.1 — Terminal: ICC run of the best design [SS-P09] . . . 15

Figure 5.2 — Post-place QoR report of the best design [SS-P10] . . . 17

Figure 5.3 — Post-place power report [SS-P11] . . . 17

Figure 5.4 — Placement utilisation report [SS-P12] . . . 17

Figure 5.5 — IC Compiler layout window of the placed best design [SS-P13] . . . 17

Figure 5.6 — Block outline, port boxes and power straps (cells hidden) [SS-P13b] . . . 17

Figure 5.7 — Placed standard cells between the power straps [SS-P13c] . . . 18

Figure 5.8 — Core-size sweep summary [SS-P14] . . . 18

Figure 6.1 — PlaceOpt run listing (17 runs) [SS-P15] . . . 19

Figure 7.1 — Best design before (DC) and after (ICC PlaceOpt) placement . . . 20

List of Tables

Table 2.1 — Tools and technology actually used . . . 7
Table 2.2 — External delay used per iteration (0.4 × clock period) . . . 7
Table 3.1 — Variables of the DC driver script . . . 8
Table 4.1 — Table 5: Post-Synthesis QoR Data (Design Compiler, normal synthesis) . . . 11
Table 4.2 — All clock periods run (DC and ICC): timing, size, power and area . . . 12
Table 4.3 — Iteration log with the quantified improvement . . . 13
Table 4.4 — Derived quantities (formula, then value) . . . 14
Table 5.1 — Candidate designs and the reason for the decision . . . 14
Table 5.2 — ICC place-and-route settings . . . 15
Table 5.3 — Table 6: Post-Place QoR Data (IC Compiler, after PlaceOpt) . . . 16
Table 5.4 — Complete data of the best design (r4h12, 1.20 ns) . . . 16
Table 5.5 — PlaceOpt runs of r4h12 and r4h11 at different core sizes (all clean) . . . 18
Table 6.1 — All PlaceOpt runs on the server (17): periods, timing, size, power . . . 19
Table 7.1 — Verification of the PD numbers and files . . . 20
Table 7.2 — Compliance summary for the physical-design requirements . . . 20
Table 8.1 — Problem log . . . 21
Table A.1 — Commands of the physical-design work . . . 22
Table B.1 — Screenshot checklist . . . 23
Table D.1 — Traceability . . . 24
Table E.1 — Abbreviations . . . 24

Chapter 1 Introduction

1.1 Background and problem statement

The course project asks for a hardware QR-code decoder written in Bluespec SystemVerilog. The decoder, `mkQRDecoder`, receives the side length of a symbol, reads the dark/light modules through a one-module-per-cycle memory port, corrects damage (BCH(15,5) on the format bits and Reed–Solomon over GF(2^8) on the data blocks) and returns the message bytes. Beyond a correct design, the course requires that the design is taken through physical design tools and that its power, performance and area (PPA) are improved by analysing synthesis reports and re-designing (spec Section 6.2).

1.2 Objectives of this report

- Document the **five synthesis iterations** (≥ 4 required) with quantified improvement, and fill in **Table 5**.
- Document the **place-and-route** of the best design with **IC Compiler** (not ICC2), and fill in **Table 6**.
- Record every command, script and configuration used so that the work can be reproduced.
- Report auxiliary experiments honestly, including those that did not lead to the best design.

1.3 Specification requirements that this report answers

SPECIFICATION SECTION 6.2 — PHYSICAL DESIGN (AS STATED IN THE ASSIGNMENT)	
Iteration (6.2.2)	Every design goes through Design → Synthesis → Analyse → Re-design; goal: improve PPA after analysing synthesis reports. Fill Table 5 after each iteration.
Synthesis rules	**40 percent of the clock period** as external delay; re-use the udcounter scripts; **do not change the timing libraries**; **Design Compiler, normal synthesis, not topographical**.
Place and route (6.2.3)	Best design only; re-use udcounter scripts; **ICC and not ICC2 or Fusion Compiler**; fill Table 6. (Staff announcement of 3 Nov: results are taken after **PlaceOpt**, utilisation ≥ 70%, Table 6 is the post-place table.)
Table footnotes	* Critical Path Clk Period and Critical Path Slack are taken from the **f2f** path group. ** Max clock frequency = clock period specified in the SDC without negative slack. Power from the **report_power** command.
Deliverable	Table values in a separate **pd_report.pdf**; folder layout `pd/design_1..4, pd/best_design` as in spec Section 6.3.
Marks	Design-space exploration 10 marks (≥ 4 post-synthesis iterations with quantifiable improvement); Place and Route of the best design 5 marks.

1.4 Division of work

The team has two members. The backend RTL (Reed–Solomon, block store), Design Compiler and IC Compiler runs, this report and the final packaging were done by Kishore (EE26E016). The reference models and the front-end/payload verification collateral were done by the partner, so that the author of a model is not the designer of the corresponding RTL (spec Section 5.2). Rishi's separate experimental 4-bank store (called R5 below) is reported only as an auxiliary experiment.

1.5 Organisation of the report

Chapter 2 gives the tools and set-up; Chapter 3 the synthesis flow and commands; Chapter 4 the five iterations and Table 5 with the full timing table; Chapter 5 place-and-route and Table 6 with the best-design data sheet; Chapter 6 the auxiliary experiments; Chapter 7 the analysis; Chapter 8 the problems met; Chapter 9 the conclusion. The appendices hold the command reference, the screenshot index, the folder inventory, the traceability matrix and the glossary.

Chapter 2 Tools, Libraries and Environment

2.1 Software and technology

Table 2.1 — Tools and technology actually used

Item	Value	Used for
Server	varaha-eda (IP 10.21.227.38), working tree `/scratch/CS6230_EE26E016/`	All DC / ICC runs
HDL compiler	Bluespec `bsc` 2021.12.1 (build fd501401)	BSV → Verilog
Synthesis	Synopsys Design Compiler U-2022.12 (`dc_shell`), normal (non-topographical) mode, wire-load model	Table 5
Place & route	Synopsys IC Compiler V-2023.12 (`icc_shell`) — not ICC2	Table 6
Technology	SAED 14nm EDK 03_2025, RVT standard cells	Library
Setup corner	`saed14rvt_base_ss0p72v125c.db` (0.72 V, 125 °C)	DC and ICC setup
Hold corner (ICC)	ff0p88v125c (0.88 V, 125 °C)	ICC hold analysis
Synthetic libraries	`dw_foundation.sldb`, `standard.sldb`	DesignWare

The timing libraries are those of the udcounter example and were **not changed**, as the specification requires. The decoder RTL is generated by `bsc` and analysed as Verilog together with the Bluespec library files (`FIFO2.v, RegFile.v, SizedFIFO.v`).

2.2 Constraints (SDC)

The SDC is identical in every run apart from the period *P*. Input and output delays are set to 40% of the period on every non-clock port:

```
create_clock -name CLK -period <P> -waveform {0.0 <P/2>} [get_ports CLK]
set_input_delay  -clock CLK <0.4*P> [all_inputs]
set_output_delay -clock CLK <0.4*P> [all_outputs]
```

Table 2.2 — External delay used per iteration (0.4 × clock period)

Design	Clock period (ns)	Input / output delay (ns)
1	5.00	2.00
2	5.00	2.00
3	2.50	1.00
4	2.00	0.80
5	1.20	0.48



Figure 2.1 — The SDC file and the library set-up of the DC script [SS-P01]

Chapter 3 Synthesis Flow with Design Compiler

3.1 Flow overview

Each iteration follows the same steps: (1) generate Verilog from the Bluespec source with the options of that iteration; (2) run the team DC script (`run_dc_full.sh`, derived from the udcounter scripts) with the clock period and cell restrictions of the iteration; (3) read the QoR, timing, power and area reports; (4) decide the next change. The script is driven by environment variables:

Table 3.1 — Variables of the DC driver script

Variable	Meaning	Example
TAG	name of the run; reports go to reports_<TAG>, netlist to outputs_<TAG>	r4h12
PERIOD	clock period in ns	1.2
GATE	1 = enable clock gating in compile_ultra	1
BSC_FLAGS	bsc `-D` options that select the RTL variant	-D QR_R4_BM -D QR_STORE_PIPE
DONT_USE	library cells DC must not use	SAEDRVT14_OAI22_* ...

3.2 Commands used for the DC runs

The driver generates Verilog with exactly these course flags (verified from the script):

```
bsc -u -verilog -vdir . -bdir . -info-dir . -remove-dollar $FLAGS -g mkQRDecoder QRDecoder.bsv ▫
    +RTS -M16g -RTS
```

Reconstructed invocations. The script interface and the option names are verified from the script text; the exact shell lines typed for each of the five runs are reconstructed from the run names and the compile options recorded in the submission README, and must be confirmed against the shell history (they are therefore marked *reconstructed* in Appendix A).

```
# Design 1 (r0_plain, 5.0 ns)   [reconstructed]
TAG=r0_plain PERIOD=5.0 GATE=0 BSC_FLAGS="-D QR_R0_PLAIN" bash run_dc_full.sh
# Design 2 (r2, 5.0 ns)         [reconstructed]
TAG=r2 PERIOD=5.0 GATE=1 BSC_FLAGS="-D QR_R2_BM" bash run_dc_full.sh
# Design 3 (r4_2p5, 2.5 ns)     [reconstructed]
TAG=r4_2p5 PERIOD=2.5 GATE=1 BSC_FLAGS="-D QR_R4_BM -D QR_STORE_PIPE" bash run_dc_full.sh
# Design 4 (r4g20, 2.0 ns)      [reconstructed]
TAG=r4g20 PERIOD=2.0 GATE=1 DONT_USE="SAEDRVT14_OAI22_* SAEDRVT14_OAI21_* SAEDRVT14_ND2_CDC_* ▫
    SAEDRVT14_MUX2_MM_4" bash run_dc_full.sh
# Design 5 / best_design (r4h12, 1.2 ns)   [reconstructed]
TAG=r4h12 PERIOD=1.2 GATE=1 DONT_USE="SAEDRVT14_OAI22_* SAEDRVT14_OAI21_* SAEDRVT14_ND2_CDC_* ▫
    SAEDRVT14_MUX2_MM_4" bash run_dc_full.sh
```

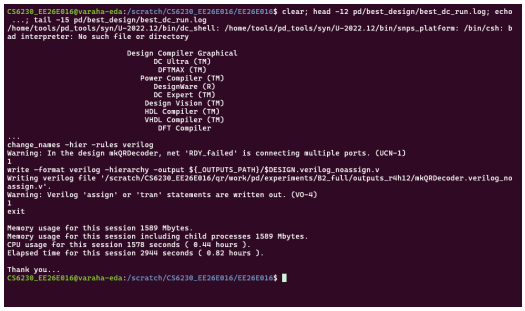


Figure 3.1 — Terminal: the DC run of the best design (r4h12, 1.2 ns) [SS-P02]

The first line of the log ("/bin/csh: bad interpreter") is a start-up warning of the `dc_shell` wrapper on this server; synthesis ran normally and wrote the netlist (elapsed 2,944 s). The warning about net RDY_failed connecting two ports is a normal naming warning of `change_names`.

3.3 Reports produced per run

For every design the following files are in the submission under `pd/design_N/` (best design: `pd/best_design/`): `*_dc_qor.rpt`, `*_dc_timing.rpt`, `*_dc_power.rpt`, `*_dc.tcl`, `*_dc.sdc`, `dc_run.log`. Table 5 values come from the QoR report (f2f group, leaf-cell and sequential-cell counts, design area) and from `report_power`.

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; grep -n -i -E "Critical Path Length|Critical
Path Slack|Leaf Cell Count|Sequential Cell Count|Cell Area|Design Area|Total Negative Slack|Violating" pd/design_1
/design_1_dc_qor.rpt | head -40
15:  Critical Path Length:          4.98
16:  Critical Path Slack:           0.00
18:  Total Negative Slack:          0.00
19:  No. of Violating Paths:        0.00
28:  Critical Path Length:          0.73
29:  Critical Path Slack:           2.27
31:  Total Negative Slack:          0.00
32:  No. of Violating Paths:        0.00
41:  Critical Path Length:          2.95
42:  Critical Path Slack:           0.00
44:  Total Negative Slack:          0.00
45:  No. of Violating Paths:        0.00
56:  Leaf Cell Count:             115224
62:  Sequential Cell Count:        32102
77:  Cell Area:              60138.156456
78:  Design Area:           216341.522279
103:  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
106:  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 3.2 — Design 1: QoR report [SS-P03]

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; grep -n -i -E "Critical Path Length|Critical
Path Slack|Leaf Cell Count|Sequential Cell Count|Cell Area|Design Area|Violating" pd/design_2/design_2_dc_qor.rpt
| head -40
15:  Critical Path Length:          4.09
16:  Critical Path Slack:           0.00
19:  No. of Violating Paths:        0.00
28:  Critical Path Length:          4.98
29:  Critical Path Slack:           0.00
32:  No. of Violating Paths:        0.00
41:  Critical Path Length:          0.47
42:  Critical Path Slack:           2.53
45:  No. of Violating Paths:        0.00
54:  Critical Path Length:          2.91
55:  Critical Path Slack:           0.00
58:  No. of Violating Paths:        0.00
69:  Leaf Cell Count:              90815
75:  Sequential Cell Count:        36247
90:  Cell Area:              53818.216319
91:  Design Area:           141711.740854
116:  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
119:  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 3.3 — Design 2: QoR report [SS-P04]

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; grep -n -i -E "Critical Path Length|Critical
Path Slack|Leaf Cell Count|Sequential Cell Count|Cell Area|Design Area|Violating" pd/design_3/design_3_dc_qor.rpt
| head -40
15:  Critical Path Length:          2.11
16:  Critical Path Slack:           0.00
19:  No. of Violating Paths:        0.00
28:  Critical Path Length:          2.48
29:  Critical Path Slack:           0.00
32:  No. of Violating Paths:        0.00
41:  Critical Path Length:          0.51
42:  Critical Path Slack:           0.99
45:  No. of Violating Paths:        0.00
54:  Critical Path Length:          1.45
55:  Critical Path Slack:           0.00
58:  No. of Violating Paths:        0.00
69:  Leaf Cell Count:              92463
75:  Sequential Cell Count:        35744
90:  Cell Area:              55068.209499
91:  Design Area:           140112.704854
116:  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
119:  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 3.4 — Design 3: QoR report [SS-P05]

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; grep -n -i -E "Critical Path Length|Critical
Path Slack|Leaf Cell Count|Sequential Cell Count|Cell Area|Design Area|Violating" pd/design_4/design_4_dc_qor.rpt
| head -40
15:  Critical Path Length:          1.76
16:  Critical Path Slack:           0.00
19:  No. of Violating Paths:        0.00
28:  Critical Path Length:          1.97
29:  Critical Path Slack:           0.00
32:  No. of Violating Paths:        0.00
41:  Critical Path Length:          0.52
42:  Critical Path Slack:           0.68
45:  No. of Violating Paths:        0.00
54:  Critical Path Length:          1.17
55:  Critical Path Slack:           0.00
58:  No. of Violating Paths:        0.00
69:  Leaf Cell Count:              89407
75:  Sequential Cell Count:        36657
90:  Cell Area:              53701.488735
91:  Design Area:           151986.890066
116:  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
119:  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 3.5 — Design 4: QoR report [SS-P06]

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; grep -n -i -E "Critical Path Length|Critical
Path Slack|Leaf Cell Count|Sequential Cell Count|Cell Area|Design Area|Violating" pd/best_design/best_design_dc_qo
r.rpt | head -40
15:  Critical Path Length:          0.51
16:  Critical Path Slack:           0.00
19:  No. of Violating Paths:        0.00
28:  Critical Path Length:          1.19
29:  Critical Path Slack:           0.00
32:  No. of Violating Paths:        0.00
41:  Critical Path Length:          0.40
42:  Critical Path Slack:           0.32
45:  No. of Violating Paths:        0.00
54:  Critical Path Length:          0.69
55:  Critical Path Slack:           0.00
58:  No. of Violating Paths:        0.00
69:  Leaf Cell Count:              92233
75:  Sequential Cell Count:        36667
90:  Cell Area:              55777.543858
91:  Design Area:           143448.015110
116:  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
119:  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 3.6 — Design 5 (best): QoR report [SS-P07]

```
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$ clear; grep -n -i -B3 -A3 "Total" pd/best_design/bes
t_design_dc_power.rpt | tail -25
41-  Cell Internal Power  =   6.4960 mW   (33%)
42-  Net Switching Power  =  13.2605 mW   (67%)
43-                         ---------
44:Total Dynamic Power    =  19.7566 mW  (100%)
45-
46-Cell Leakage Power     = 131.9894 uW
47-
48-
49:                  Internal         Switching           Leakage            Total
50-Power Group       Power            Power               Power              Power   (   %    )  Attrs
51--------------------------------------------------------------------------------------------------
52-io_pad             0.0000           0.0000              0.0000             0.0000  (   0.00%)
--
57-sequential         0.0000           0.0000              0.0000             0.0000  (   0.00%)
58-combinational      0.2788           1.6354          4.8200e+07             1.9623  (   9.87%)
59--------------------------------------------------------------------------------------------------
60:Total              6.4965 mW        13.2605 mW      1.3199e+08 pW         19.8907 mW
61-1
CS6230_EE26E016@varaha-eda:/scratch/CS6230_EE26E016/EE26E016$
```

Figure 3.7 — Design 5: report_power [SS-P08]

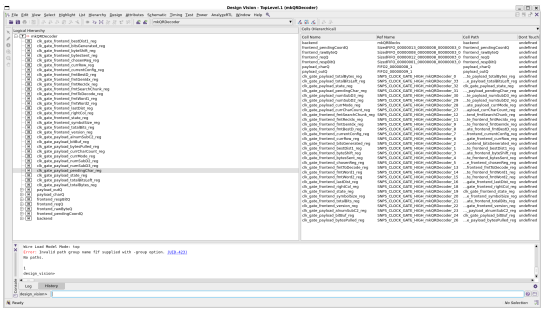


Figure 3.8 — Design Vision: logical hierarchy of mkQRDecoder (r4h12) [SS-P16]

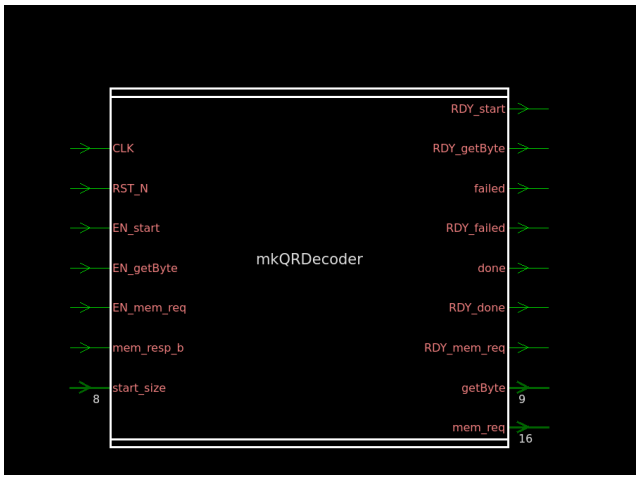


Figure 3.9 — Design Vision: top-level schematic symbol of mkQRDecoder [SS-P17]

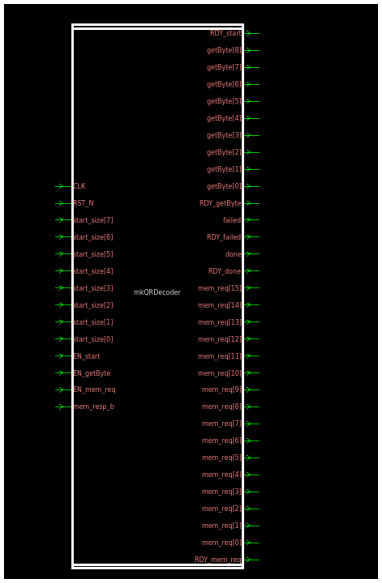


Figure 3.10 — Design Vision: schematic with all bus bits expanded [SS-P17b]

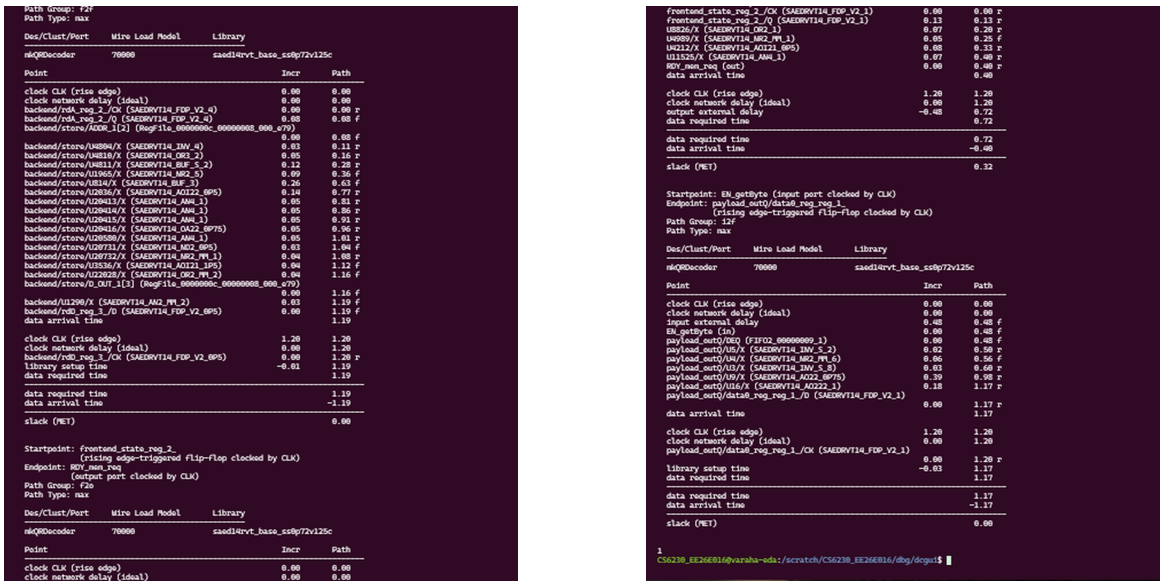


Figure 3.11 — Design Vision: report_timing on the saved DC result [SS-P18]

Chapter 4 Post-Synthesis Results: Design-Space Exploration (Table 5)

4.1 Table 5 — Post-synthesis QoR data (official)

RESULT — TABLE 5 SUMMARY	
Clock achieved	**5.0 ns → 1.2 ns** (200 MHz → **833.3 MHz**), 4.17× faster, 0 violating paths in all five runs
f2f critical path	**4.98 ns → 1.19 ns**
Cell area	**60,138.2 → 55,777.5 µm² (-7.3%)**
Design area (DC)	**216,341.52 → 143,448.02 µm² (-33.7%)**
Leaf cells	**115,224 → 92,233 (-20.0%)**
Power at equal 5.0 ns	**9.9178 → 4.5249 mW (-54.4%)** (Design 1 → Design 2)

Table 4.1 — Table 5: Post-Synthesis QoR Data (Design Compiler, normal synthesis)

Design Number	Critical Path Clk Period* (ns)	Critical Path Slack* (ns)	Max Clock Freq** (MHz)	Total Number of Leaf Cells	Total number of sequential cells	Total power (mW)	Design Area (µm²)
1	5.00	0.00	200.0	115,224	32,102	9.9178	216,341.52
2	5.00	0.00	200.0	90,815	36,247	4.5249	141,711.74
3	2.50	0.00	400.0	92,463	35,744	8.8798	140,112.70
4	2.00	0.00	500.0	89,407	36,657	11.9537	151,986.89
5 (best)	1.20	0.00	833.3	92,233	36,667	19.8907	143,448.02

* From the f2f path group of the QoR report; slack 0.00 means met (f2f Critical Path Length: 4.98, 4.98, 2.48, 1.97, 1.19 ns). ** The clock period written in the SDC, with no negative slack. Power is the Total row of `report_power`. Design 5 is also the `pd/best_design` folder. Design area is the DC “Design Area” (cells plus estimated interconnect).

4.2 Complete timing table — every clock period run (ns)

The table below lists **every** clock period that was run through synthesis or placement, including auxiliary runs that are not part of Table 5, so that the timing behaviour of each architecture across periods can be read in one place. “Source” says whether the numbers were read from the server reports by the author or taken from the partner's report.

Table 4.2 — All clock periods run (DC and ICC): timing, size, power and area

Design / variant	Tool	Clock period (ns)	f2f path (ns)	Slack, violating paths	Fmax (MHz)	Leaf / seq. cells	Power (mW)	Area (µm²)	Source
Design 1 (r0_plain)	DC	5.00	4.98	met, 0	200.0	115,224 / 32,102	9.9178	216,341.52 (cell 60,138.2)	server
Design 1 (r0_plain)	DC	3.00	2.95	met, 0	333.3	118,631 / 32,102	17.2569	213,584.9 (cell 61,720.1)	server
Design 1 (r0_plain)	DC	2.00	2.78	−0.80, >500 viol.	— (fails)	120,717 / 32,102	24.4300	—	partner report
Design 2 (r2)	DC	5.00	4.98	met, 0	200.0	90,815 / 36,247	4.5249	141,711.74 (cell 53,818.2)	server
Design 3 (r4_2p5)	DC	2.50	2.48	met, 0	400.0	92,463 / 35,744	8.8798	140,112.70 (cell 55,068.2)	server
Design 4 (r4g20)	DC	2.00	1.97	met, 0	500.0	89,407 / 36,657	11.9537	151,986.89 (cell 53,701.5)	server
Design 5 (r4h12)	DC	**1.20**	**1.19**	**met, 0**	**833.3**	**92,233 / 36,667**	**19.8907**	**143,448.02** (cell 55,777.5)	server
r4h11	DC	1.10	1.08	met, 0	909.1	93,536 / 36,667	21.6118	144,886.2 (cell 56,331.4)	partner report
r4h12 placed (best_design)	ICC PlaceOpt	**1.20**	**0.80**	**0.00, 0**	**833.3**	**85,490 / 36,667**	**7.3985**	**52,285.97** (util 92.29%)	server
R5 (4-bank, experimental)	DC	0.70	0.69	met, 0	1428.6	92,267 / 37,020	10.1773	137,036.54 (cell 57,385.1)	server
R5 (4-bank, experimental)	DC	0.60	—	met, 0	1666.7	—	11.8705	140,700.47 (cell 59,673.11)	server
R5 (4-bank, experimental)	DC	0.55	0.54	met, 0	1818.2	97,050 / 37,020	13.0061	140,392.13 (cell 60,719.66)	server
r4h11 placed, core 244 µm	ICC PlaceOpt	1.10	0.75	0.00, 0	909.1	86,823 / 36,667	7.9398 (dyn.)	52,626.12 (util 92.89%)	server
R5 placed (netlist 0.60), core 254 µm	ICC PlaceOpt	0.60	0.58	0.00 as printed, 8 f2f viol.	— (not clean)	87,555 / 37,021	9.0973 (dyn.)	53,365.29 (util 86.97%)	server
R5 placed (netlist 0.55), core 256 µm	ICC PlaceOpt	0.55	0.53	0.00 as printed, 5 f2f viol.	— (not clean)	87,254 / 37,020	9.9689 (dyn.)	53,739.54 (util 86.27%)	server

Dashes mean the value was not recorded in the reports reviewed. Runs *r4h13*, *r4h15* and an *R1 3.0 ns* run also exist on the server; their numbers are not needed for any conclusion and are not tabulated. DC power and ICC power are **not comparable** (different tools, wire models and switching-activity defaults); see Chapter 7.

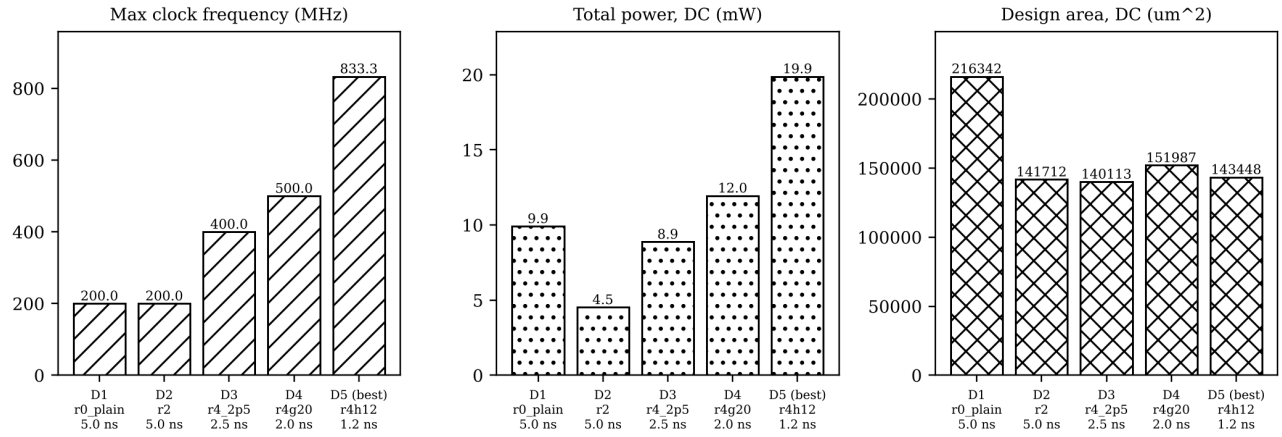


Figure 4.1 — Table 5 results: frequency, DC power and DC design area for the five iterations

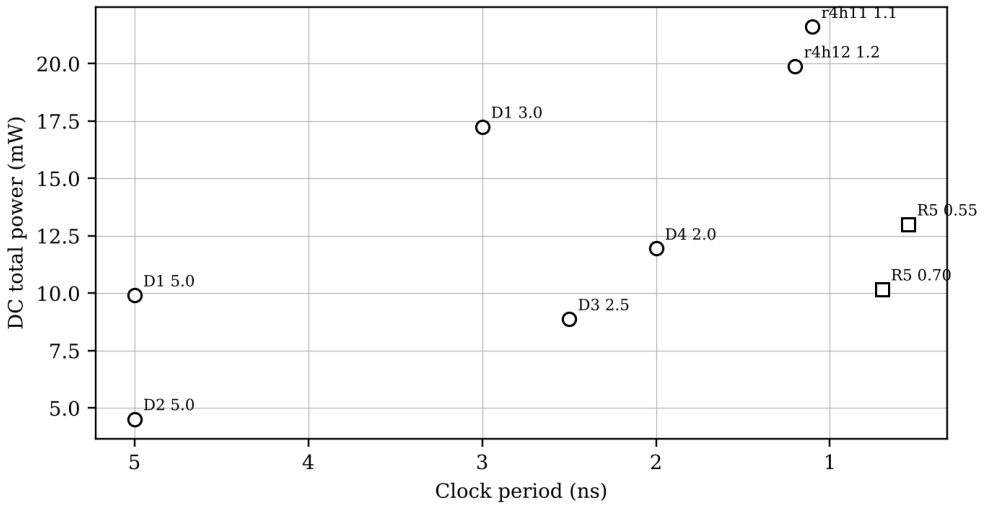


Figure 4.2 — DC total power against clock period for every passing run (circles: full decoder variants; squares: experimental R5)

4.3 What changed in each iteration (Design → Synthesis → Analyse → Re-design)

Table 4.3 — Iteration log with the quantified improvement

Iteration	Analysis of previous reports	Re-design applied	Quantified result
1 (r0_plain)	Baseline decoder: single-cycle Berlekamp–Massey (BM) loop, un-gated 29.6 kbit codeword store.	— (baseline)	5.00 ns met; cell area 60,138.2; power 9.9178 mW; 158 logic levels (partner report).
2 (r2)	Power report: almost all power in un-gated register clock pins and recirculating muxes of the store; BM cone dominates combinational area.	BM split into 3 phases (`-D QR_R2_BM`), clock gating on (GATE=1), same 5.0 ns to isolate the effect.	Power 9.9178 → 4.5249 mW (-54.4%); design area -34.5%; cell area -10.5%.
3 (r4_2p5)	With power fixed, the critical path (BM and store access) limits the clock.	2-stage BM (`-D QR_R4_BM`) and 1-cycle store pipeline (`-D QR_STORE_PIPE`); clock 2.5 ns.	2.5 ns met (2× faster than iteration 2); cell area +2.3% vs iteration 2.
4 (r4g20)	Remaining long paths: divisions (/3, /45, /100) in front end and payload; library cells with pin-access problems.	Divider-free reciprocal multiply-shift, pipelined BCH search and zigzag, cells excluded with dont_use; 2.0 ns.	2.0 ns met; cell area 53701.5 (smallest of all five); 17 logic levels (partner report).
5 (r4h12, best)	Design 4 closes with margin on all path groups.	Same architecture, further timing fixes to the backend; clock tightened to 1.2 ns.	1.2 ns met, f2f 1.19 ns; 4.17× faster than iteration 1; cell area -7.3% vs iteration 1.

EXPLANATION OF THE POWER TREND

Power rises from 4.52 mW (5.0 ns) to 19.89 mW (1.2 ns) because dynamic power scales with clock frequency and DC upsizes cells to meet the tighter period. The comparison that isolates architecture from frequency is iteration 1 → 2 at the identical 5.0 ns period: **-54.4% power**.

4.4 Derived metrics: formulas and values

Table 4.4 — Derived quantities (formula, then value)

Quantity	Formula	Value
Maximum frequency	f = 1000 / T[ns] MHz	5.0 → 200.0; 2.5 → 400.0; 2.0 → 500.0; 1.2 → 833.3
Clock speed-up D1 → D5	5.00 / 1.20	4.17×
Cell-area change D1 → D5	(55,777.5 − 60,138.2) / 60,138.2	-7.25%
Design-area change D1 → D5	(143,448.02 − 216,341.52) / 216,341.52	-33.69%
Power change D1 → D2 (5.0 ns)	(4.5249 − 9.9178) / 9.9178	-54.38%
Leaf-cell change D1 → D5	(92,233 − 115,224) / 115,224	-19.95%
External delay	0.4 × T	1.2 ns → 0.48 ns

Chapter 5 Place and Route of the Best Design (Table 6)

5.1 Choice of the best design

The best design is the *r4h12* netlist at 1.20 ns. It has the highest clock frequency of the independently developed RTL, closes DC with zero violating paths and, as Section 5.4 shows, places cleanly over a window of core sizes. The faster experimental R5 netlists (down to 0.55 ns in DC) were also placed and did **not** close timing (Chapter 6), and R5 is not independent of the model author, so it was not selected.

5.2 Every candidate period down to 0.55 ns and why it was or was not chosen

Table 5.1 — Candidate designs and the reason for the decision

Candidate	Period (ns)	Stage reached / result	Decision and reason
Design 1 r0_plain	5.00	DC met; 9.9178 mW; cell 60,138.2	Not chosen: baseline; slowest clock, largest area and highest power at 5 ns.
Design 1 r0_plain	3.00	DC met (f2f 2.95); 17.2569 mW	Not chosen: limit of the un-gated baseline; +74% power for a 1.7× faster clock.
Design 1 r0_plain	2.00	DC fails, slack −0.80, > 500 violating paths	Not usable: the baseline architecture cannot reach 2 ns.
Design 2 r2	5.00	DC met; 4.5249 mW (lowest of all)	Not chosen: lowest power but 4.2× slower than the best design; the clock was the lever still available.
Design 3 r4_2p5	2.50	DC met; 8.8798 mW	Superseded: the same family closes at 1.2 ns.
Design 4 r4g20	2.00	DC met; 11.9537 mW	Superseded: closes with margin, so the period could be tightened further.
Design 5 r4h12	**1.20**	**DC met; placed clean (0 setup / 0 hold) at core 242, 244, 246 µm**	**CHOSEN**: Table 5 design 5 with the complete evidence set; lowest placed power and smallest placed area of the clean candidates; f2f margin 0.40 ns after placement; clean in 3 of 3 core sizes tried.
r4h11	1.10	DC met (f2f 1.08); 21.6118 mW; **placed clean** at core 242 and 244 (0 violating paths; f2f 0.75 ns)	**Viable alternative, not chosen**: 9.1% higher frequency than r4h12 but 9% more dynamic power after placement (7.9398 vs 7.2825 mW), 0.65% more placed area (52,626.12 vs 52,285.97 µm²), a smaller f2f margin (0.35 vs 0.40 ns) and two core sizes tried instead of three. The energy per clock cycle is equal (about 8.7 pJ), so the choice is frequency against power and margin.
R5 4-bank	0.70	DC met; 10.1773 mW; area 137,036.54	Not chosen: placed at 0.70 ns, the R5 netlist left 1 f2f violation (core 250) and 2 (core 254).
R5 4-bank	0.60	DC met; 11.8705 mW; cell 59,673.11	Not chosen: 3 placements at 0.60 ns left 8, 8 and 9 f2f violations; at 0.65 ns, 4 and 5.
R5 4-bank	0.55	DC met (f2f 0.54); 13.0061 mW; cell 60,719.66	Not chosen: 5 placements at 0.55 ns left 4, 5, 5, 12 and 14 f2f violations.

REASONS R5 WAS NOT SELECTED

(1) **Timing**: none of the 12 R5 placements (periods 0.55 to 0.70 ns, cores 250–262 µm) was clean: 1 to 14 f2f setup violations of a few picoseconds. (2) **Area and power after placement**: 52,948.78–53,844.41 µm² and 7.83–10.00 mW (dynamic) against 52,285.97 µm² and 7.2825 mW dynamic for r4h12. (3) **Independence**: R5's RTL and its model were written by the same person, which the spec warns against (Section 5.2); the final design must be the one whose RTL and model were separated. (4) A clean 1.2 ns result with a margin window is safer than a sub-ns result that depends on a lucky placement run.

5.3 Floorplan and flow settings

Table 5.2 — ICC place-and-route settings

Item	Setting
Tool / stage	icc_shell V-2023.12, stop after PlaceOpt (`QR_STOP=place`); CTS and routing not run (staff announcement 3 Nov)
Netlist	`outputs_r4h12/mkQRDecoder.verilog_noassign.v` from Design Compiler
Constraints	same SDC as DC: 1.2 ns clock, 0.48 ns input / output delay; ideal clock
Floorplan	`create_floorplan -control_type width_and_height -core_width L -core_height L`, L = CORE_L = 244 µm
Power straps	M4 / M5 / M6 power straps
Placement	`set_place_opt_strategy -layer_optimization_effort high`, `-consider_routing true`; `place_opt -effort high -congestion -continue_on_missing_scandef`
Cells excluded	SAEDRVT14_OAI22_*, OAI21_*, ND2_CDC_*, MUX2_MM_4 (`set_dont_use`, same list as DC); 0 such cells in the placed netlist
Corners	setup: ss0p72v125c; hold: ff0p88v125c

5.4 Commands used for place and route

```
cd /scratch/CS6230_EE26E016/qr/work/pd/icc
QR_STOP=place TAG=r4h12 PERIOD=1.2 CORE_L=244 NAME=pn_r4h12_L244 \
  DONT_USE="SAEDRVT14_OAI22_* SAEDRVT14_OAI21_* SAEDRVT14_ND2_CDC_* SAEDRVT14_MUX2_MM_4" \
  nohup bash run_icc.sh > pn_r4h12_L244.out 2>&1 &
```

The invocation above is the one recorded for the final run (`pn_r4h12_L244`). `run_icc.sh` looks up the DC netlist and QoR report of run TAG, writes the SDC for PERIOD, and starts icc_shell with the scripts in `pd/icc/scripts/` (`icc_initial_setup.tcl, icc_lib_setup.tcl, icc_qr.tcl`).

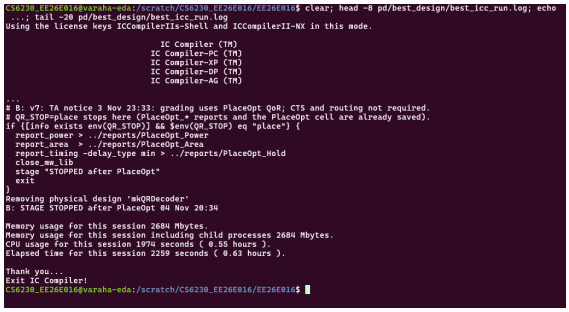


Figure 5.1 — Terminal: ICC run of the best design [SS-P09]

5.5 Table 6 — Post-place QoR data (official)

RESULT — TABLE 6 BEST DESIGN AFTER PLACEOPT	
Timing	**0 violating setup paths, 0 violating hold paths**; clock 1.20 ns (833.3 MHz); f2f critical path 0.80 ns
Utilisation	**92.29%** (requirement ≥ 70%)
Block dimensions	**243.98 × 243.45 µm**
Placed cell area	**52,285.97 µm²**
Total power (report_power)	**7.3985 mW**
Leaf / sequential cells	**85,490 / 36,667**

Table 5.3 — Table 6: Post-Place QoR Data (IC Compiler, after PlaceOpt)

Design Number	Critical Path Clk Period* (ns)	Critical Path Slack* (ns)	Max Clock Freq** (MHz)	Total Number of Leaf Cells	Total number of sequential cells	Total power (mW)	Block Dimensions of the Layout (µm)	Design Area (µm²)
1 (best_design = r4h12)	1.20	0.00	833.3	85,490	36,667	7.3985	243.98 × 243.45	52,285.97

* From the f2f group of `best_design_post_place_qor.rpt`. ** 1000/1.20 ns. Power from `best_design_post_place_power.rpt` (`report_power`). Design area is the placed standard-cell area (no net area exists before routing), so it is comparable with the DC *cell* area (55,777.5 µm²), not the DC *design* area (143,448.02 µm²).

5.6 Best-design data sheet (all details)

Table 5.4 — Complete data of the best design (r4h12, 1.20 ns)

Parameter	Post-synthesis (DC)	Post-place (ICC PlaceOpt)
Netlist / tag	r4h12	pn_r4h12_L244 (CORE_L = 244)
Clock period / frequency	1.20 ns / 833.3 MHz	1.20 ns / 833.3 MHz
External I/O delay	0.48 ns (40%)	0.48 ns (40%)
Critical path (f2f)	1.19 ns (16 logic levels)	0.80 ns (7 logic levels)
Setup / hold violating paths	0 / —	0 / 0 (worst hold slack +0.02 ns)
Other path groups (placed)	i2f 0.69 ns, f2o 0.40 ns	i2f 0.45 ns (slack 0.00), f2o 0.62 ns (slack +0.10)
Total leaf cells	92,233	85,490
Sequential cells	36,667	36,667
Cell area	55,777.5 µm²	52,285.97 µm² (comb. 19,286.47 + non-comb. 32,999.50)
Design area	143,448.02 µm² (incl. wire estimate)	= cell area (placement only)
Core / block size	—	243.98 × 243.45 µm, core area 59,396.92 µm², 387 rows
Utilisation	—	92.29% (1,177,619 of 1,275,939 sites)
Total power	19.8907 mW	7.3985 mW
Power breakdown	—	internal 6.3528 + switching 0.9290 + leakage 0.1167 mW
Wire model / clock	wire-load model 70000, ideal clock	placement parasitics, ideal clock
Forbidden cells (dont_use)	excluded	0 in the placed netlist

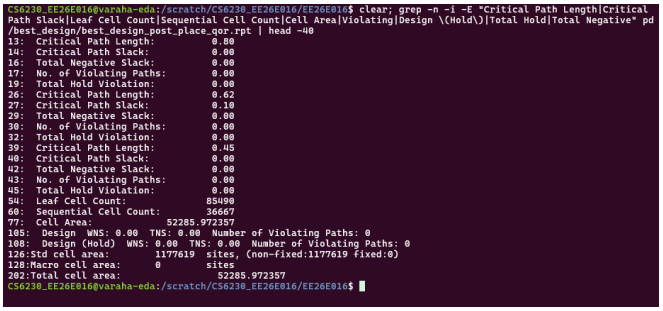


Figure 5.2 — Post-place QoR report of the best design [SS-P10]

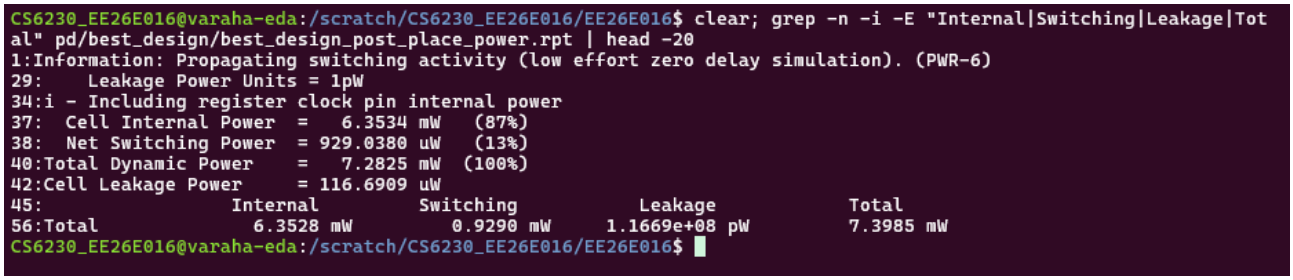


Figure 5.3 — Post-place power report [SS-P11]

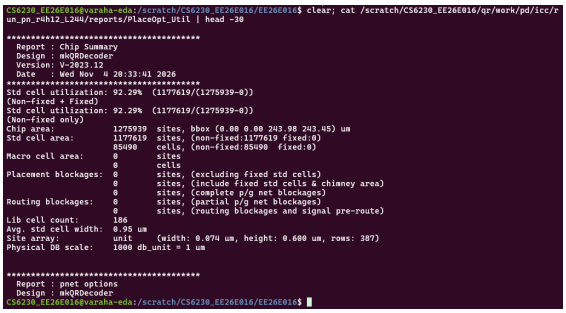


Figure 5.4 — Placement utilisation report [SS-P12]

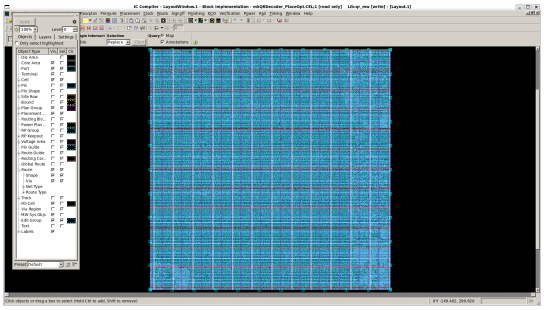


Figure 5.5 — IC Compiler layout window of the placed best design [SS-P13]

The screenshot above is the layout window of the IC Compiler layout window (opened read-only on the saved PlaceOpt cell with `icc_shell -shared_license -gui`). The block is the 243.98 × 243.45 µm core; the horizontal and vertical stripes are the M4/M5/M6 power straps and the rows between them are filled with placed standard cells (92.29% utilisation). No routing or clock tree exists at this stage, so the picture shows placement and power planning only.

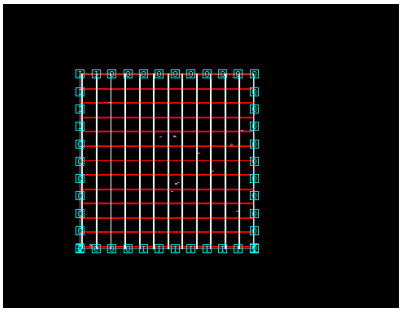


Figure 5.6 — Block outline, port boxes and power straps (cells hidden) [SS-P13b]

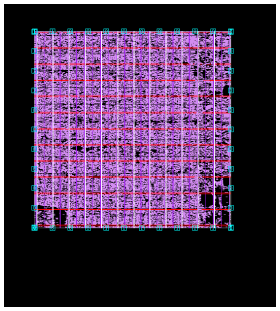


Figure 5.7 — Placed standard cells between the power straps [SS-P13c]

5.7 Core-size robustness of the chosen design

ICC is not bit-for-bit repeatable, so a single clean run could be luck. The r4h12 netlist was therefore placed at several core sizes; all of them closed with zero violating paths:

Table 5.5 — PlaceOpt runs of r4h12 and r4h11 at different core sizes (all clean)

Netlist	Core side L (µm)	f2f path (ns)	Violating paths	Leaf cells	Cell area (µm²)	Utilisation	Dynamic power (mW)
r4h12 (1.20 ns)	242	0.82	0	85,501	52,269.68	93.75%	7.2827
r4h12 (1.20 ns), submitted	**244**	**0.80**	**0**	**85,490**	**52,285.97**	**92.29%**	**7.2825**
r4h12 (1.20 ns)	246	0.81	0	85,484	52,262.04	90.57%	7.3302
r4h11 (1.10 ns)	242	0.76	0	86,816	52,636.24	94.41%	7.9557
r4h11 (1.10 ns)	244	0.75	0	86,823	52,626.12	92.89%	7.9398

Dynamic power is the Total Dynamic Power row of the PlaceOpt power report (the submitted run's Total row is 7.3985 mW). The chosen size lies inside a passing window and not at an edge; the 1.10 ns netlist r4h11 also places clean. Only the 244 µm run is the submitted one, and its full report set is in `pd/best_design/`.

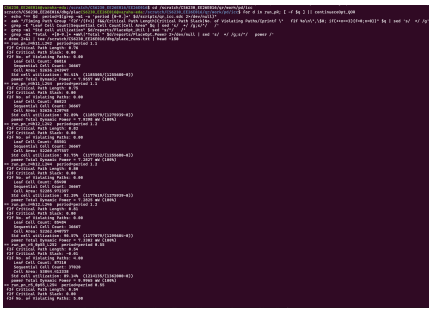


Figure 5.8 — Core-size sweep summary [SS-P14]

Chapter 6 Auxiliary Experiments

6.1 Frequency sweeps of earlier designs

Design 1 (un-gated baseline) was also pushed: it closes at 3.00 ns (f2f path 2.95 ns) but at a power cost (17.2569 mW against 9.9178 mW at 5.00 ns), and fails at 2.00 ns with −0.80 ns slack. The r4h12 architecture closes down to 1.10 ns in DC (variant r4h11, f2f path 1.08 ns) and that netlist also places clean (Chapter 5). These runs show that the improvement of Table 5 is architectural and not an artefact of the chosen periods.

6.2 R5: a 4-bank store (experimental)

The partner built a different backend (called R5) that splits the codeword store into four banks and uses bit-matrix GF multipliers. It is functionally correct (240/240 hardware cases in the full-decoder sweeps) and closes DC at 0.70 ns (10.1773 mW), 0.60 ns (11.8705 mW) and 0.55 ns (13.0061 mW). It was then placed in **12 PlaceOpt runs** (netlist r5_0p55 at 0.55 ns; netlist r5_0p60 at 0.60, 0.65 and 0.70 ns; cores 250–262 µm). **No run was clean**: every run had 1 to 14 f2f setup violations of a few picoseconds. *(An earlier draft of the partner report said 13 runs and quoted other values for two runs; the table below gives the values read from the server reports.)*

Table 6.1 — All PlaceOpt runs on the server (17): periods, timing, size, power

Run (netlist, ICC period, core L)	f2f path (ns)	f2f slack (ns)	Violating f2f paths	Leaf cells	Seq. cells	Cell area (µm²)	Util.	Dyn. power (mW)
r4h11, 1.10, 242	0.76	0.00	0	86,816	36,667	52,636.24	94.41%	7.9557
r4h11, 1.10, 244	0.75	0.00	0	86,823	36,667	52,626.12	92.89%	7.9398
r4h12, 1.20, 242	0.82	0.00	0	85,501	36,667	52,269.68	93.75%	7.2827
r4h12, 1.20, 244 (submitted)	**0.80**	**0.00**	**0**	**85,490**	**36,667**	**52,285.97**	**92.29%**	**7.2825**
r4h12, 1.20, 246	0.81	0.00	0	85,484	36,667	52,262.04	90.57%	7.3302
R5 0p55, 0.55, 252	0.54	−0.01	4	87,310	37,020	53,844.41	89.14%	9.9965
R5 0p55, 0.55, 254	0.54	0.00	5	87,140	37,020	53,690.88	87.53%	9.9997
R5 0p55, 0.55, 256	0.53	0.00	5	87,254	37,020	53,739.54	86.27%	9.9689
R5 0p55, 0.55, 258	0.54	−0.01	12	87,204	37,020	53,740.34	84.78%	10.0039
R5 0p55, 0.55, 262	0.53	0.00	14	87,253	37,020	53,803.52	82.57%	9.9599
R5 0p60, 0.60, 250	0.52	0.00	8	87,505	37,021	53,326.89	89.85%	9.1556
R5 0p60, 0.60, 254	0.58	0.00	8	87,555	37,021	53,365.29	86.97%	9.0973
R5 0p60, 0.60, 258	0.58	0.00	9	87,570	37,021	53,413.64	84.23%	9.1309
R5 0p60, 0.65, 250	0.63	0.00	4	—	—	—	—	8.4371
R5 0p60, 0.65, 254	0.64	0.00	5	87,427	37,021	53,088.86	86.51%	8.3887
R5 0p60, 0.70, 250	0.68	0.00	1	87,367	37,021	52,948.78	89.21%	7.8292
R5 0p60, 0.70, 254	0.68	0.00	2	87,352	37,021	—	—	—

Slack is shown as printed in the report (rounded to 0.01 ns, so “0.00” with violating paths means a few picoseconds negative). A dash means the value was not visible in the recorded output of the summary loop. Dyn. power is the Total Dynamic Power line of the PlaceOpt power report. The 7.3985 mW of Table 6 is the Total row of the same report for the submitted run, which also includes leakage; dynamic values of different runs are compared with each other only.

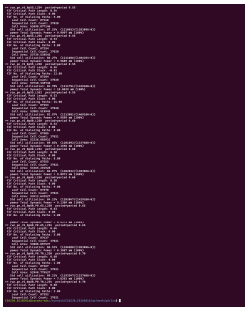


Figure 6.1 — PlaceOpt run listing (17 runs) [SS-P15]

Chapter 7 Analysis and Discussion

7.1 DC versus ICC for the same netlist

The best design shows fewer leaf cells, less area and less power after placement than after synthesis. This is not a placement miracle; it is a difference of method:

- DC uses a wire-load model (70000) that is pessimistic for a block of this size and adds buffering; ICC places real cells and removes part of it, so the leaf count falls from 92,233 to 85,490 (-7.3%).
- DC “Design Area” includes estimated interconnect area (143,448.02 µm²); ICC after PlaceOpt reports standard-cell area only (52,285.97 µm²). Compare 52,285.97 with the DC cell area 55,777.5 (-6.3%).
- Total power is 7.3985 mW (ICC) against 19.8907 mW (DC). The tools use different default switching-activity assumptions, and ICC at this stage has an ideal clock and no routing parasitics. The two power values must not be read as a power saving.

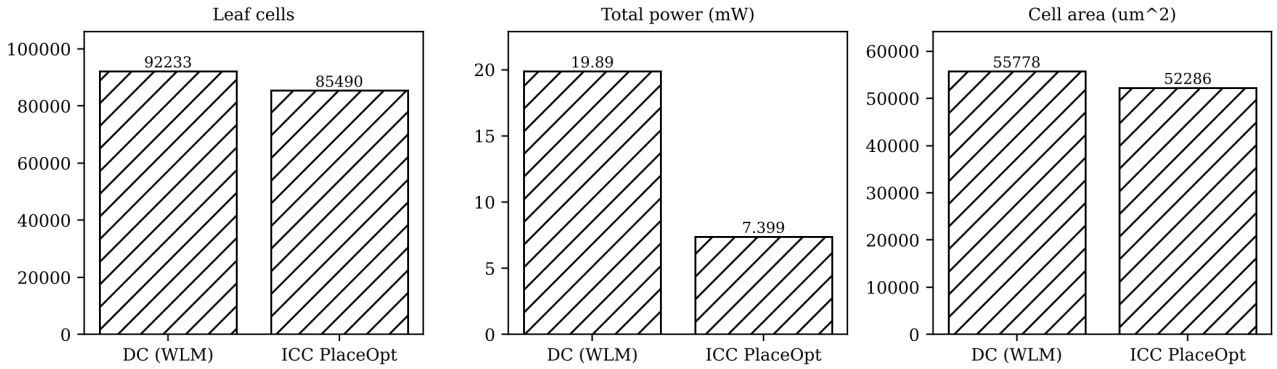


Figure 7.1 — Best design before (DC) and after (ICC PlaceOpt) placement

7.2 Findings of the design-space exploration

- Power: clock gating plus the 3-phase BM cut power by 54.4% at the same period (iteration 2).
- Performance: pipelining BM and the store, then removing dividers and pipelining the front end, allowed the period to fall from 5.0 ns to 1.2 ns; the sequential-cell count stays near 36,000.
- Area: the standard-cell area is almost flat (53.7–55.8 thousand μm²) once the design is gated, so frequency was gained at little area cost.
- Trade-off: the fastest clock needs the most power in DC, as expected.

7.3 Checks performed on the physical-design results

Table 7.1 — Verification of the PD numbers and files

Check	Method	Outcome
Table 5 numbers	re-read from each server dc_qor / dc_power report (leaf, sequential, power, design area, f2f length)	all five rows agree with the reports
Slack and violating paths	QoR report of every DC run	slack 0.00, 0 violating paths in all five runs
Table 6 numbers	re-read from post-place QoR, power and utilisation reports	agree; utilisation 92.29%
Hold timing after placement	ff0p88v125c corner	0 violating paths, worst hold slack +0.02 ns
Forbidden cells	search of the placed netlist for the dont_use patterns	0 found
Core-size robustness	r4h12 placed at 242, 244, 246 μm (r4h11 at 242, 244)	all 0 violating paths
Repeatability	ICC runs are not bit-for-bit repeatable	documented; the submitted run is the one whose reports are in best_design/
Typos in the partner draft	compared against server reports	power of designs 3 and 4 corrected; R5 run count corrected to 12
Rules of Section 6.2	SDC 0.4 × T, normal DC, ICC not ICC2, libraries unchanged	all satisfied (Table 7.2)
Functional equivalence of the netlists	the DC netlists come from the same bsv/ sources whose md5 sums are listed in report.pdf	same RTL as the verified design; gate-level simulation was **not** run

7.4 Compliance with the specification

Table 7.2 — Compliance summary for the physical-design requirements

Requirement	Status	Evidence
≥ 4 post-synthesis iterations with quantifiable improvement	Met (5)	Table 5; Section 4.3
40% clock-period external delay	Met	SDC in every run; Table 2.2
DC normal (non-topographical) synthesis	Met	dc_shell U-2022.12, wire-load model, no topographical mode
Timing libraries unchanged	Met	saed14rvt_base_ss0p72v125c.db as in the udcounter example
ICC (not ICC2) for the best design	Met	icc_shell V-2023.12
Utilisation ≥ 70% (staff announcement)	Met	92.29%
Table 5 and Table 6 filled with real numbers	Met	Chapters 4 and 5
DRC / LVS clean layout (6.2.3 text)	**Not demonstrated**	Per the staff announcement the flow stops after PlaceOpt; routing, DRC and LVS were not run.

Chapter 8 Problems Met and How They Were Resolved

Table 8.1 — Problem log

Problem	Cause	Resolution
ICC flow not repeatable run to run	multi-threaded placement; small timing differences	placed r4h12 at several core sizes; submitted a run that is clean and checked that neighbours are clean too
Pin-access / spacing trouble with a few library cells	cells OAI22_*, OAI21_*, ND2_CDC_*, MUX2_MM_4	excluded with dont_use in DC and ICC (iteration 4 onward)
Table 5 power values mistyped in the partner draft (8.8785, 11.9544)	copy error	corrected to 8.8798 and 11.9537 from the server reports
R5 placement count quoted as 13	miscount	corrected to 12
DC and ICC numbers confused as comparable	different area/power definitions	explained in Section 5.4 and Chapter 7
Report file names for the post-route table	staff changed the stage to PlaceOpt	files named `*_post_place_*`, Table 6 titled Post-Place QoR Data

A security review of the final RTL found: only the course ports, no system tasks, only library cells, and the payload length bounds-checked; the codeword store is not cleared between symbols (not observable at the interface).

Chapter 9 Conclusion and Future Work

FINAL RESULTS

Table 5: five iterations, 5.00 ns → 1.20 ns (200 → 833.3 MHz, 4.17×), f2f path 4.98 → 1.19 ns, cell area -7.3%, 0 violating paths in every run. **Table 6:** best design (r4h12, 1.20 ns) after ICC PlaceOpt: 0 setup / 0 hold violations, 92.29% utilisation, 243.98 × 243.45 µm, 52,285.97 µm², 7.3985 mW.

The exploration met the requirement of at least four post-synthesis iterations with quantified improvement, and the best design was placed with IC Compiler above the utilisation threshold. Limitations: the layout was not routed (stage defined by the staff announcement), so DRC/LVS cleanliness is not demonstrated; power numbers use default activity.

Future work: clock-tree synthesis and routing with DRC/LVS signoff; activity-annotated power; re-balancing the R5 bank crossbar so that a sub-nanosecond period also places cleanly.

Appendix A Command Reference

Status key: **verified** = text of the project script or a recorded run; **reconstructed** = rebuilt from script interface and run names, to be confirmed against shell history.

Table A.1 — Commands of the physical-design work

ID	Command	Directory	Purpose	Status
CMD-P01	ssh / scp to the course server (IP 10.21.227.38)	PC	Access the server and upload / download files	verified
CMD-P02	bsc -u -verilog -vdir . -bdir . -info-dir . -remove-dollar $FLAGS -g mkQRDecoder QRDecoder.bsv	build_<TAG>/	BSV to Verilog	verified (script)
CMD-P03	TAG=r0_plain PERIOD=5.0 GATE=0 BSC_FLAGS="-D QR_R0_PLAIN" bash run_dc_full.sh	pd/experiments/B2_full	Design 1 synthesis	reconstructed
CMD-P04	TAG=r2 PERIOD=5.0 GATE=1 BSC_FLAGS="-D QR_R2_BM" bash run_dc_full.sh	same	Design 2 synthesis	reconstructed
CMD-P05	TAG=r4_2p5 PERIOD=2.5 GATE=1 BSC_FLAGS="-D QR_R4_BM -D QR_STORE_PIPE" bash run_dc_full.sh	same	Design 3 synthesis	reconstructed
CMD-P06	TAG=r4g20 PERIOD=2.0 GATE=1 DONT_USE="..." bash run_dc_full.sh	same	Design 4 synthesis	reconstructed
CMD-P07	TAG=r4h12 PERIOD=1.2 GATE=1 DONT_USE="..." bash run_dc_full.sh	same	Design 5 / best synthesis	reconstructed
CMD-P08	dc_shell (called by run_dc_full.sh with dc.tcl and the SDC)	work_<TAG>/	compile_ultra, reports	verified (script)
CMD-P09	QR_STOP=place TAG=r4h12 PERIOD=1.2 CORE_L=244 NAME=pn_r4h12_L244 DONT_USE="..." nohup bash run_icc.sh	pd/icc	ICC floorplan + PlaceOpt of the best design	verified
CMD-P10	icc_shell (called by run_icc.sh with icc_qr.tcl)	run_<NAME>/work	floorplan, place_opt, reports	verified (script)
CMD-P11	bash icc_summary.sh	pd/icc	collect violating-path counts of the runs	verified (script exists)
CMD-P12	cp reports to pd/best_design/ with *_post_place_* names; bash pack2.sh	EE26E016/	assemble and zip the submission	verified
CMD-P13	md5sum bsv/*.bsv; zip -r EE26E016.zip EE26E016	server	integrity check and package	verified

Appendix B Screenshot Evidence Index

All screenshots are to be saved in `/mnt/c/Users/KOTAPATI KISHORE/Documents/CS6230_screenshots/` under the file names below; rebuilding the report with the images in place inserts them automatically at the marked slot.

Table B.1 — Screenshot checklist

ID	File name	Content	Section	Need	Status
SS-P01	SS-P01_sdc_and_dc_script.png	The SDC file and the library set-up of the DC script	2.2	required	inserted
SS-P02	SS-P02_dc_command_design5.png	Terminal: the DC run of the best design (r4h12, 1.2 ns)	3.2	required	inserted
SS-P03	SS-P03_design1_qor.png	Design 1: QoR report	3.3	required	inserted
SS-P04	SS-P04_design2_qor.png	Design 2: QoR report	3.3	required	inserted
SS-P05	SS-P05_design3_qor.png	Design 3: QoR report	3.3	required	inserted
SS-P06	SS-P06_design4_qor.png	Design 4: QoR report	3.3	required	inserted
SS-P07	SS-P07_design5_qor.png	Design 5 (best): QoR report	3.3	required	inserted
SS-P08	SS-P08_design5_power.png	Design 5: report_power	3.3	required	inserted
SS-P16	SS-P16_dc_hierarchy.png	Design Vision: logical hierarchy of mkQRDecoder (r4h12)	3.3	optional	inserted
SS-P17	SS-P17_dc_schematic.png	Design Vision: top-level schematic symbol of mkQRDecoder	3.3	optional	inserted
SS-P17b	SS-P17b_dc_schematic_ports.png	Design Vision: schematic with all bus bits expanded	3.3	optional	inserted
SS-P18	SS-P18_dc_timing_report.png	Design Vision: report_timing on the saved DC result	3.3	optional	inserted
SS-P09	SS-P09_icc_command_and_log.png	Terminal: ICC run of the best design	5.3	required	inserted
SS-P10	SS-P10_post_place_qor.png	Post-place QoR report of the best design	5.5	required	inserted
SS-P11	SS-P11_post_place_power.png	Post-place power report	5.5	required	inserted
SS-P12	SS-P12_utilization.png	Placement utilisation report	5.5	required	inserted
SS-P13	SS-P13_layout_gui.png	IC Compiler layout window of the placed best design	5.5	optional	inserted
SS-P13b	SS-P13b_floorplan_straps.png	Block outline, port boxes and power straps (cells hidden)	5.5	optional	inserted
SS-P13c	SS-P13c_placed_cells.png	Placed standard cells between the power straps	5.5	optional	inserted
SS-P14	SS-P14_core_sweep_summary.png	Core-size sweep summary	5.6	optional	inserted
SS-P15	SS-P15_r5_place_summary.png	PlaceOpt run listing (17 runs)	6.2	optional	inserted

Appendix C Submission Folder Inventory (pd/)

```
EE26E016/pd/
  design_1/  design_1_dc_qor.rpt  design_1_dc_timing.rpt  design_1_dc_power.rpt ▫
      design_1_dc.tcl  design_1_dc.sdc  dc_run.log
  design_2/  ... design_3/ ... design_4/ ...   (same six files each)
  best_design/
     best_design_dc_qor.rpt  best_design_dc_timing.rpt  best_design_dc_power.rpt ▫
         best_design_dc.tcl  best_design_dc.sdc  best_dc_run.log
     best_design_post_place_qor.rpt  best_design_post_place_power.rpt  best_design_place.tcl ▫
         best_design_place.sdc  best_icc_run.log
     icc_support/   (ICC set-up scripts needed to re-run)
```

Appendix D Requirement Traceability Matrix (physical design)

Table D.1 — Traceability

Req.	Requirement	Where implemented	Verification	Result	Evidence
PD-1	≥ 4 DC iterations, quantified	pd/design_1..4, best_design	Table 5	5 iterations	Ch. 4
PD-2	40% external delay	*_dc.sdc	SDC inspection	0.4 × T	Table 2.2
PD-3	Normal DC synthesis	*_dc.tcl	no topographical options	yes	Ch. 3
PD-4	Libraries unchanged	dc.tcl / icc_lib_setup.tcl	same .db as example	yes	Table 2.1
PD-5	ICC for best design	pd/best_design/*_place.tcl	icc_shell log	yes	SS-P09
PD-6	Table 6 (post-place) filled	post_place reports	report numbers	Table 6	Ch. 5
PD-7	Utilisation ≥ 70%	floorplan L = 244	utilisation report	92.29%	SS-P12
PD-8	Table 5 from f2f group, report_power	QoR + power reports	reports	yes	SS-P03..P08
PD-9	Layout DRC/LVS clean	—	not run (stage ends at PlaceOpt)	not demonstrated	—

Appendix E Glossary and Abbreviations

Table E.1 — Abbreviations

Term	Meaning
BM	Berlekamp–Massey key-equation solver
BCH	Bose–Chaudhuri–Hocquenghem code (format information)
DC	Synopsys Design Compiler
DSE	Design-space exploration
f2f / i2f / f2o	flop-to-flop / input-to-flop / flop-to-output path groups
ICC	Synopsys IC Compiler
PPA	Power, performance, area
QoR	Quality of results
RVT	Regular threshold voltage cell library
SAED14	Synopsys 14nm educational PDK
SDC	Synopsys design constraints
WLM	Wire-load model
WNS / TNS	Worst / total negative slack
PlaceOpt	ICC place_opt placement and optimisation step

Appendix F References

- [1] CS6230 CAD for VLSI, *Project: QR Code Decoder* (specification), released 15 Oct 2026, including Section 6 and Tables 5–7.
- [2] Synopsys, *Design Compiler* U-2022.12 user documentation (as installed on the course server).
- [3] Synopsys, *IC Compiler* V-2023.12 user documentation (as installed on the course server).
- [4] Synopsys, SAED 14nm EDK (03_2025) standard-cell libraries and the udcounter example scripts supplied by the course.
- [5] Bluespec, Inc., *bsc* 2021.12.1 compiler and BSV reference guide.
- [6] ISO/IEC 18004, *QR Code bar code symbology specification* (background only).